

# Dart Throwing in Virtual Reality with Aim Assistance

Ren Boz\*  
University of Arizona

## ABSTRACT

Throwing tasks in virtual reality (VR) are often challenging due to limitations in depth perception, force control, and the absence of familiar physical feedback. Aim assistance may help alleviate some of these challenges, potentially allowing users to perform throwing actions more accurately and consistently. This demo presents the application of aim assistance in a VR dart-throwing task. The demonstrated system incorporates a subtle, trajectory-based aim assistance method that operates at the moment of release. Rather than visibly guiding the trajectory, our approach gently adjusts the dart's initial flight direction to partially compensate for throwing errors. During the demo, attendees can directly experience and compare dart throwing with and without aim assistance, allowing them to observe the differences in accuracy, control, and the overall interaction experience firsthand.

**Index terms:** Virtual reality, aim assistance, dart throwing.

## 1 INTRODUCTION AND BACKGROUND

Virtual reality (VR) enables users to engage with immersive, interactive environments that approximate real-world activities across entertainment, training, and rehabilitation contexts [1]. Despite the advances in hand tracking, handheld controllers remain the dominant means of interaction for most VR systems [2]. Throwing is a fundamental motor skill that plays a key role in sports, games, and therapeutic exercises, and it is frequently incorporated into VR applications. However, prior work has shown that throwing performance in VR often falls short of real-world performance, with reduced accuracy and consistency [3, 4]. These issues are commonly attributed to imperfect depth cues, limited or no proprioceptive feedback, and differences between virtual and physical object dynamics. Nevertheless, throwing remains a common component of VR applications, including training and rehabilitation scenarios [5, 6], as it is a natural and easily understood motor action. Aim assistance techniques have been successfully applied in video gaming contexts to improve accuracy and user experience [7, 8]. Within extended reality environments, such techniques have also been associated with benefits such as improved performance [9], lower perceived pressure [10], and increased engagement [11]. Nonetheless, there is limited empirical understanding of how aim assistance affects user experience in immersive VR throwing tasks, motivating the work demonstrated here.

\* e-mail: rboz@arizona.edu

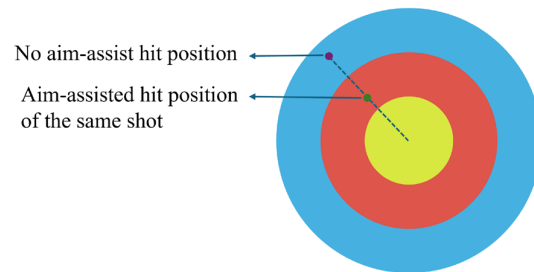


Figure 1: Visualization of the aim-assistance correction. The figure illustrates the relationship between the unassisted and assisted dart trajectories. The predicted landing location based solely on the user's input is indicated by a red mark, while the corrected landing position after aim assistance is shown in green.

## 2 IMPLEMENTATION

The demo allows users to perform a dart-throwing task in VR under conditions with and without aim assistance, highlighting its effects on both performance and user experience. The system employs an invisible bullet-magnetism aim assistance approach that modifies the dart's trajectory at the instant of release. Specifically, the initial throwing direction is adjusted such that the deviation between the dart's landing point and the center of the target is reduced by 50% along the line connecting these points (Fig. 1). To prevent the assistance from becoming noticeable during highly inaccurate throws, the correction is applied only when the dart is released within a limited projected distance to the center of the target.

Users interact with a handheld Meta Quest 2 controller through a trigger-based grasp-and-release interaction metaphor, aiming by rotating the controller and throwing darts by releasing the trigger while swinging the controller forward. Pulling the trigger on the dominant-hand controller spawns a virtual dart directly into the user's hand, avoiding the need for explicit object selection via direct manipulation. Once spawned, the dart is spatially attached to the controller and visually aligned between the thumb, index, and middle fingers until thrown.

To throw, users perform a forward swinging motion with their dominant hand and release the trigger. The system estimates the throw direction and speed by averaging the velocity vectors of the controller over the final five sampled poses immediately preceding the trigger release, sampled at 0.02-second intervals. The resulting velocity is then instantaneously applied to the virtual dart in the computed direction (Fig. 2).

Dart motion is simulated using Unity's physics engine, incorporating standard gravitational acceleration ( $9.81 \text{ m/s}^2$ ) along the negative y-axis, with no linear or angular drag. To reduce ambiguity in release timing and mitigate frustration associated with incorrect release timing, the implementation uses a short trigger travel distance. Based on internal testing, the dart is grasped when the trigger exceeds 0.95 of its normalized range and released once it falls below 0.90, where 0 represents a fully released trigger and 1 a fully depressed trigger. This design choice reduces uncertainty at



Figure 2: Dart-throwing interaction sequence. The figure presents screenshots from the VR task illustrating the throwing. Users aim and throw using a handheld controller by releasing the trigger while swinging the controller forward. A trajectory preview is shown during aiming, followed by a brief trajectory trail rendered immediately after the release. When the dart strikes the board, the impact location is highlighted, and a score notification is displayed.

release and accidental drops, thereby supporting a stronger sense of control during throwing.

Upon impact, the landing position is marked on the dartboard, and a score indicator overlay appears (Fig. 2). The target is divided into three concentric scoring zones with diameters of 0.2 m (inner), 0.4 m (middle), and 0.6 m (outer). The center of the dartboard is positioned 1.73 m above the floor, and users stand at a fixed distance of 2.37 m from the dartboard, maintained by disabling forward locomotion via real-world movements (Fig. 3).

### 3 CONCLUSION

This demo provides an experiential illustration of how subtle, trajectory-based aim assistance affects throwing performance in immersive VR without overtly altering interaction mechanics. By allowing attendees to directly compare assisted and unassisted conditions, the demonstration highlights the potential of aim assistance as a design approach for supporting usability and engagement in VR motor skill tasks.

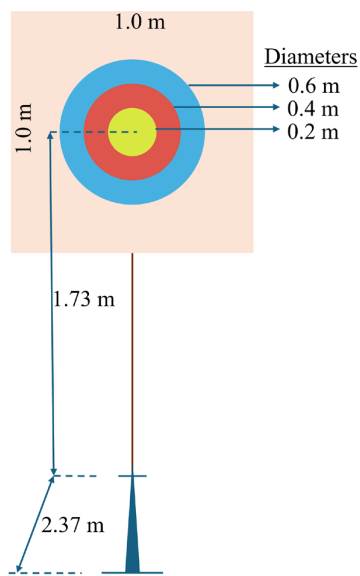


Figure 3: Schematic of the VR dart-throwing setup. The figure depicts the virtual dartboard and participant configuration used in the task. Users stand 2.37 m away from the dartboard, and the center of the dartboard is positioned 1.73 m above the floor.

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